Chapter 1
Toward Problem-Agnostic Spatial Reasoning Under Fog of War: A Hybrid Fuzzy Modular Architecture for Kriegspiel Chess

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Abstract K

riegspiel, the imperfect-information variant of chess, poses a challenging testbed for decision-making under hidden state, sparse feedback, and fog of war. Existing approaches, including metaposition-based search and Monte Carlo Tree Search (MCTS), achieve competent defensive play but remain largely reactive and tightly coupled to the specific structure of Kriegspiel [1, 2].

We propose a hybrid fuzzy–neural architecture that learns relational spatial structure, patterns of coordination, threat, and control. Evaluations against established baselines indicate that the model achieves strong positional outcomes despite limitations in endgame resolution. This work takes a step toward imperfect-information game agents that reason strategically rather than relying solely on probabilistic rollout or heuristic search.

Key words: kriegspiel, chess, fog-of-war, fuzzy logic, explainable AI, interpretable AI.

1.1 Introduction

Kriegspiel originated in 19th-century Prussia as a military wargame designed to train strategic reasoning under uncertainty. Unlike perfect-information games such as standard chess, Kriegspiel introduces an umpire who alone has full visibility of the game state. Players submit move attempts, which the umpire executes while providing only limited feedback, forcing reasoning under conditions commonly described as fog of war (FoW) [3].

In 1899, this framework was adapted into a chess variant, typically implemented using separate player boards and an umpire board that maintains the true state, as illustrated in Fig. 1.1. While multiple variants have since emerged, the Internet Chess

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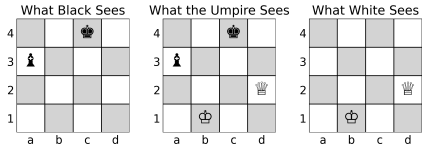

Fig. 1.1 Example of who sees what during a game of Kriegspiel Chess

Club (ICC) Wild 16 ruleset has become the de-facto competitive standard [3]. Under ICC rules, the umpire issues constrained announcements—such as checks, captures, or illegal moves—without revealing full board information. As a result, players must actively probe and infer the opponent's hidden configuration, often using exploratory or deliberately illegal moves to gain information about defended regions of the board.

Kriegspiel substantially increases the complexity of chess by replacing a single known board configuration with a belief space of all states consistent with limited umpire feedback [3]. Rather than planning from a fixed position, players must reason over many plausible hidden arrangements of opponent pieces, causing an exponential expansion of the effective search space and rendering brute-force simulation impractical without state reduction or sampling strategies.

Gameplay dynamics further diverge from standard chess. Because captures are among the few events that reveal precise opponent information, players often recapture immediately when possible, resulting in slower, more defensive play where pieces have limited high-impact actions before being removed. Existing computational approaches largely reflect this dynamic, often prioritizing survival over proactive, coordinated attacks [3].

Unlike other fog-of-war domains such as real-time strategy games [4, 5], Kriegspiel operates over discrete, instantaneous actions with no continuous movement or prolonged engagements. Consequently, individual moves are disproportionately high impact, and strategic positioning is acutely sensitive to uncertainty, making Kriegspiel a distilled testbed for macro-level decision-making under partial observability.

1.1.1 Motivation

Despite recent advances using metapositions [3], Monte Carlo Tree Search (MCTS) [2][6], and related imperfect-information algorithms, most Kriegspiel engines display a strong defensive bias. They react to discovered threats but rarely construct attacking formations or long-horizon plans. In other words, they often survive and luck into winning positions rather than truly playing the game with an understanding of the game. This motivates the central problem addressed in this work: how to design a Kriegspiel engine, under ICC rules, that can learn relational spatial structure, patterns in how pieces co-locate, coordinate, threaten, and control space, so it

can proactively build advantageous positions and infer hidden state, with the goal of outperforming rollout-heavy or heuristic search baselines which tend towards very problem/environment specific solutions to these problems which are non transferable.

1.2 Related Works

1.2.1 Metapositions [1, 3]

Ciancarini and Favini introduced the concept of metapositions as a belief-state abstraction for Kriegspiel, enabling imperfect-information play to be treated using traditional minimax-style search [1]. Their Darkboard 1.0 engine represents a set of plausible game states consistent with umpire feedback and operates directly over this abstraction without enumerating concrete positions.

Metaposition updates incorporate both player move attempts and unknown opponent actions via structured state-transition operators, while evaluation is performed at the belief-set level rather than on individual states.

This approach demonstrated robust competitive performance on ICC by enabling effective defensive play under uncertainty, but relies on hand-designed evaluation heuristics and search procedures rather than learned spatial or relational representations.

1.2.2 Monte Carlo Tree Search [2, 3]

Ciancarini and Favini later revisited Kriegspiel using Monte Carlo Tree Search (MCTS), proposing several adaptations to handle rapidly aging uncertainty [2]. Their approach restructures the search tree to reason over player moves and corresponding umpire feedback rather than explicit opponent states, enabling rollouts in perception space rather than assuming omniscient board knowledge. Among the evaluated variants, a shallow, message-driven simulation strategy combined with UCT selection consistently outperformed deeper rollout schemes and a Darkboard 1.0 minimax baseline, achieving strong results in both ICC play and competition settings.

This work demonstrates the effectiveness of adaptive, bandit-based search under severe partial observability, but remains dependent on handcrafted rollout policies and does not learn relational spatial structure across belief states.

1.2.3 "Giraffe" Chess Engine [7]

Giraffe's primary contribution lies in its structured representation of chess positions and its use of a neural network to learn high-level evaluation without handcrafted heuristics. Rather than relying on raw board encodings, Giraffe employs a semantically rich feature vector describing piece identity and location, relational properties such as attacks and mobility, and global game-state information. These features encode abstractions traditionally considered important by human experts while allowing their relative importance to be learned end-to-end by the network.

Notably, Giraffe avoids explicit linear evaluation terms (e.g., material or piece-square tables), enabling the model to approximate expert heuristics while retaining flexibility beyond hand-engineered evaluation functions.

1.2.4 Limitations of Current Work

Despite progress in Kriegspiel engines, most existing approaches rely on probabilistic rollout or heuristic search rather than genuine positional reasoning, reacting to revealed threats instead of deliberately constructing advantageous long-horizon board configurations [3, 2, 6]. This reflects a lack of internalized spatial understanding of how pieces coordinate, control space, and form strategically meaningful structures under uncertainty.

While systems such as Giraffe demonstrate that relational piece representations can enable neural models to learn positional abstractions beyond hand-crafted heuristics [**?**], these approaches—and most grid-based methods more broadly—remain tightly coupled to fixed board topologies and domain-specific assumptions. As a result, they generalize poorly to other imperfect-information or non-grid strategic environments. Addressing this limitation by learning transferable, problem-agnostic representations of strategic structure is the central motivation of the proposed approach.

1.3 Methodology

1.3.1 Model Overview

The proposed system employs a cascading hybrid architecture (Fig. 1.2) in which decision-making is localized at the level of individual pieces, allowing information relevance to be modeled differently for each entity. This design explicitly prioritizes generalization and technology transfer: the architecture is intended to remain applicable across strategic environments with varying agent types, interaction structures, and observation models.

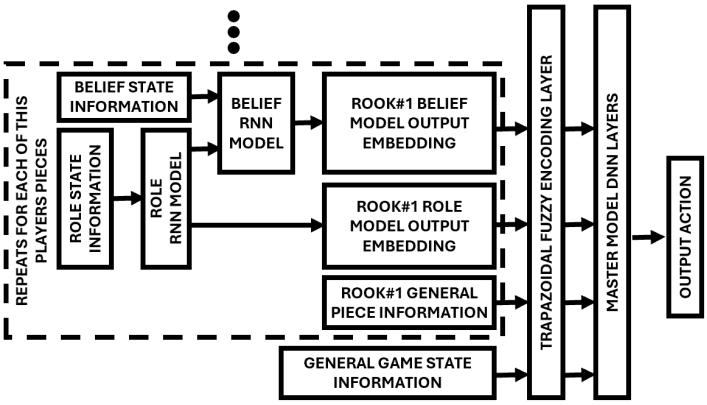

Fig. 1.2 Model architecture diagram

Consequently, the approach departs from grid-centric architectures such as convolutional neural networks (CNNs), which are well suited to rigid spatial lattices but poorly aligned with irregular, abstract, or non-Euclidean strategic domains [4, 5]. By avoiding assumptions tied to fixed board topology, the model aims to support broader applicability beyond chess-specific environments.

1.3.2 Role Model

In Kriegspiel, piece roles are asymmetric and amplified by partial observability: pawns often function as expendable information probes, while highly mobile pieces such as knights and queens exert disproportionate strategic influence [3]. To capture how a piece's functional importance evolves over time, the model learns a per-piece *role embedding*.

Role evolution is modeled using a recurrent neural network with a gated recurrent unit (GRU; hidden size 64) and a linear projection to a 32-dimensional embedding space [**?**]. The network processes an 11-dimensional normalized input encoding game phase, piece position, alive and defended status, and piece identity, enabling the embedding to represent contextual, temporally evolving strategic roles rather than isolated actions.

1.3.3 Belief Model

The belief model produces a per-piece embedding that encodes inferred state relative to the opponent's hidden configuration and allied piece interactions. By decoupling belief representation from fixed board topology or explicit opponent layouts,

the model is designed to generalize beyond chess-specific environments to broader partially observable strategic domains [3, 5].

This embedding functions as a learned pressure field, capturing inferred threat, defensive support, and tactical opportunity, and enabling decision-making based on relational belief rather than explicit opponent enumeration [2, 6].

Belief evolution over sequential umpire feedback is modeled using a GRU-based recurrent network (hidden size 64) with a linear projection to a 32-dimensional embedding [**?**]. The network processes a normalized 71-dimensional input encoding global game context, recent opponent interactions, piece-level state, and the corresponding role embedding. Together, these inputs yield a compact, temporally grounded belief representation that provides the master model with actionable strategic information under uncertainty.

1.3.4 Hybrid Fuzzy Model

The master model aggregates all available information—including global game context and the role and belief embeddings of each piece to select the final move. It is implemented as a *hybrid fuzzy deep neural network (DNN)*, designed to support smooth, interpretable decision-making under uncertainty rather than purely crisp action selection.

The first layer applies a fuzzy encoding to each input feature using three Ruspini-partitioned trapezoidal membership functions [8], passing all membership activations forward. This design reflects the inherently graded nature of strategic reasoning, where decisions are rarely binary, and aligns with prior work highlighting the benefits of fuzzy representations for reasoning under uncertainty [**?**, **?**]. Empirically, while the fuzzy-augmented model underperformed a purely neural baseline when evaluated solely on end-game win–loss outcomes, it consistently achieved lower training loss during hot-start learning, suggesting improved generalization and smoother optimization behavior (Fig. 1.3).

Following fuzzy encoding, the network consists of three fully connected hidden layers of size 1024, terminating in a move-selection layer of dimension 1,968 corresponding to a universal encoding of all possible chess moves. From the resulting logits, moves that are illegal from the player's perspective are masked to a very small value (-1e-9), after which a softmax distribution is formed. During evaluation, the move with maximum probability is selected, whereas during training, moves are sampled multinomially to encourage exploration and prevent premature policy collapse, consistent with standard reinforcement learning practice [9].

The master model operates on a fixed-length input vector of 1,194 features, combining global and per-piece information as well as the corresponding role and belief embeddings. This representation provides the model with coherent spatial and temporal context, enabling it to reason about strategic structure and piece coordination rather than relying solely on short-horizon probabilistic outcomes, a limitation observed in prior search-heavy Kriegspiel approaches [2, 6].

1.3.5 Training

Training was performed in two stages. First, the role and belief modules were independently initialized using hot-start supervised learning from approximately 17,000 recorded Kriegspiel games, then the Master model was similarly trained with the frozen role and belief model embeddings. This follows prior work showing that expert pretraining stabilizes learning in partially observable domains [**?**, **?**].

The full system was then transitioned to self-play using Temporal Difference learning against frozen and mixed opponents [**?**]. This phase did not produce improvements beyond the supervised baseline, likely due to sparse rewards and long-horizon credit assignment challenges inherent to Kriegspiel [2, 6]. Consequently, all results reported in this work are derived from the hot-start training phase, with reinforcement learning mentioned for completeness and future investigation.

1.3.6 Hot-start

All datasets were derived from the origional body of approximately 17,000 ICC games, filtered to include only player perspectives above an Elo threshold of 1850 (slightly above average player) in order to capture consistently strong play. From this process, 2,990 games met the eligibility criteria (2,691 training, 299 validation), with 1,654 games ultimately contributing to role and belief sequence generation. Dataset sizes were constrained by available hardware resources, specifically 8 GB of GPU memory on an NVIDIA Quadro RTX 4000 and 32 GB of system RAM, resulting in approximately 200k role–belief sequences and 64k master-policy sequences with a temporal horizon of 12 moves.

Role and belief models were trained using supervised sequence prediction, with targets computed directly from ground-truth game trajectories. Role targets encode outcome and activity based statistics such as survival to endgame, capture behavior, trade tendencies, normalized activity and mobility measures, and positional indicators including center presence and king defense contribution. Belief targets capture inferred tactical state, including attack pressure, defender–attacker balance, mobility and safe mobility, centrality, king zone influence, pinning, promotion potential, and multi-attack capacity, enabling the belief embedding to approximate a localized pressure field over time [3, 2].

Because ICC PGN records do not include umpire messages, these were synthetically injected during dataset generation to approximate realistic information flow. In contrast to purely random illegal-move sampling, up to two illegal move attempts were inserted between legal moves using constrained selection from currently plausible actions, ensuring informative feedback without overwhelming the sequences [3, 6].

Following hot-start training, the final prediction layers of the role and belief models were removed and the model weights frozen. The resulting 32-dimensional embeddings were used as inputs to the master model. The master policy was then hot-

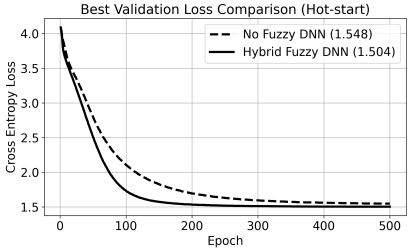

Fig. 1.3 Best validation loss in hot-start training

started using cross-entropy loss on expert move logits derived from the same high-Elo dataset. All results reported in this work are based on these hot-started models, providing a controlled evaluation of the learned embeddings and fuzzy decision architecture.

1.3.7 The Case For Fuzzy

Figure 1.3 compares validation loss during hot-start training for the Hybrid Fuzzy DNN and an otherwise identical DNN without the fuzzy encoding layer. The hybrid model converges both faster and to a lower validation loss, indicating improved optimization behavior and representational smoothness. Although full-game performance lagged behind the non-fuzzy DNN, these results suggest that with further tuning of the training procedure, the fuzzy encoding layer has the potential to outperform a purely neural baseline in end-to-end gameplay.

1.4 Results

The final trained agent, evaluated using only the hot-started model without self-play refinement, was assessed using the same methodology originally employed for the Darkboard 1.0 engine [1]. Specifically, the agent was matched against a random opponent that repeatedly sampled from its available legal move attempts until a move was accepted by the umpire.

Across 100 games, the agent achieved 24 wins, 2 losses, and 74 draws, as shown in Fig. 1.4. The draws resulted from the 55-move rule, yielding a nominal win rate of 24%. However, when drawn games were re-evaluated by assigning outcomes based on final material advantage, a common proxy for endgame dominance in chess, the agent would have been declared the winner in 81% of games and the loser in 14%.

This substantial difference would indicate that the model consistently achieved superior positions but struggled to convert material advantage into victories when

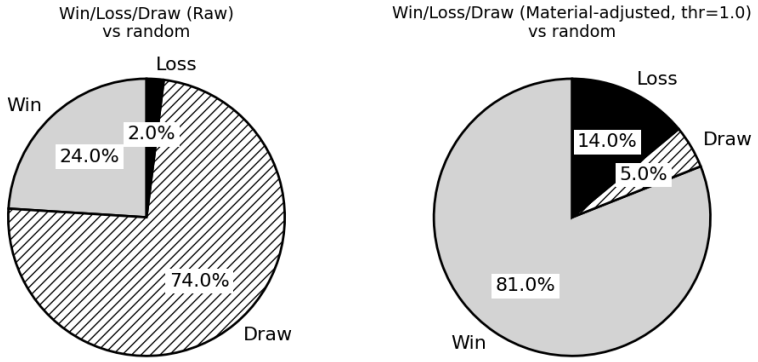

Fig. 1.4 On the left the outcome of 100 games against a random opponent, on the right the outcome if draws are re-assigned based on end game material advantage

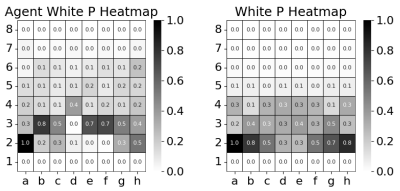

Fig. 1.5 Left agent & right ICC pawn placement

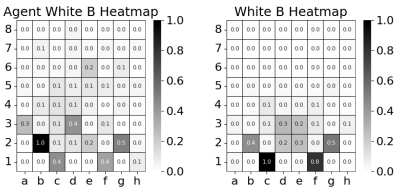

Fig. 1.6 Left agent & right ICC bishop placement

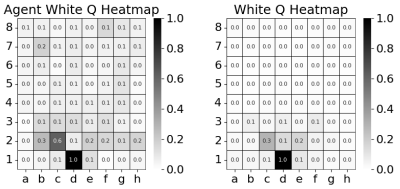

Fig. 1.7 Left agent & right ICC queen placement

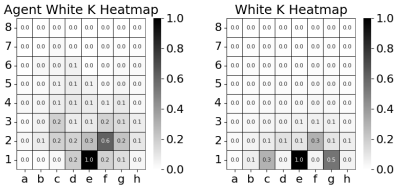

Fig. 1.8 Left agent & right ICC king placement

the endgame information sparsity began. These results suggest that while the agent demonstrates strong positional reasoning, it remains limited in endgame execution under fog of war, a hypothesis examined further in the subsequent analysis.

1.4.1 Heatmaps

We begin by comparing spatial heatmaps of piece occupancy from the 17,000-game ICC dataset against those generated by the trained agent during evaluation against a random opponent. These comparisons reveal both alignment with expert play and systematic deviations indicative of limitations in the learned embeddings.

Pawn placement (Fig. 1.5) shows partial agreement, which is expected given pawns' constrained mobility. However, the agent advances pawns toward promotion far more aggressively than observed in ICC games, where pawns are typically retained as low-value probes and recapture anchors under fog of war [3]. Over extension reduces information-gathering capacity and defensive flexibility, suggesting that pawn utility is underrepresented in the role embedding and could be improved through refined hot-start supervision.

Heatmaps for bishops and the queen (Figs. 1.6, 1.7) exhibit a shared pattern: while early-game behavior aligns with ICC data, the agent favors excessive mobility over sustained positional anchoring. In expert play, these pieces often occupy stable central-support squares that enable long-term control and coordination. The agent's tendency to relocate them frequently suggests that the belief model insufficiently encodes the strategic value of persistent central influence, prioritizing short-term activity instead.

King-related heatmaps (Fig. 1.8) reveal a more severe deficiency. The agent rarely castles and instead advances the king prematurely, increasing exposure in non-endgame positions. The observed lateral bias in queen movement further suggests compensatory behavior to shield an unsafe king. Together, these findings indicate that long-horizon king safety is not adequately captured by the current role and belief embeddings, reinforcing the need for more explicit supervision of defensive structure during hot-start training.

1.4.2 Material Advantage

Figure 1.9 provides further evidence that the agent learns consistent and effective gameplay. Across all game phases, the model maintains a clear material advantage, indicating an ability to acquire and preserve material under uncertainty. This supports the interpretation that late-game draws arise primarily from information sparsity limiting decisive action, rather than from a failure to understand or execute fundamentally strong play.

1.4.3 Agent Game-play Trends

Figure 1.10 shows the average number of illegal move attempts per game phase, stratified by outcome. Across wins and draws, illegal moves follow a roughly

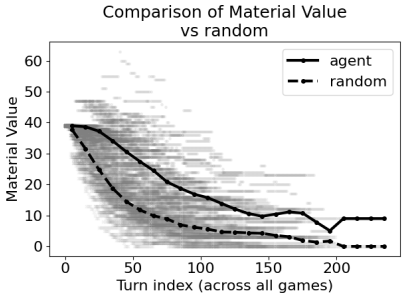

Fig. 1.9 Player material value on the board throughout a game

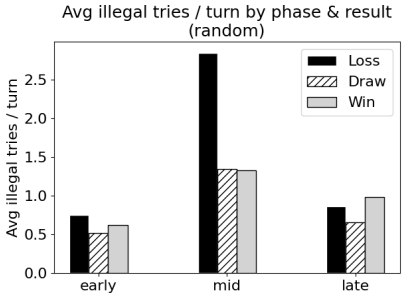

Fig. 1.10 The average number of illegal move tries compared to the game phase

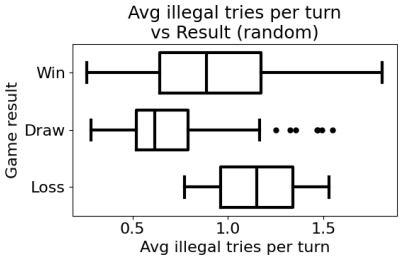

Fig. 1.11 The game result compared to average illegal tries

parabolic pattern: few attempts in the opening, a peak in the midgame as unseen opponent pieces increasingly obstruct moves, and a decline toward the endgame as material is reduced. This behavior is expected given the changing density of hidden opponent pieces over time.

Lost games exhibit a distinct deviation, with very few illegal attempts early, followed by a sharp midgame spike. One plausible explanation is that deeper-than-expected opponent infiltration in these games caused a sudden increase in obstructed moves, temporarily inflating illegal attempts before captures reduced board congestion.

Fig. 1.11 further shows that drawn games have the fewest illegal move attempts, consistent with reduced information acquisition and a higher likelihood of inconsequential play. Importantly, the model does not exhaustively probe all moves: average illegal attempts remain between 0.25 and 2.0 per turn, closely matching reported human behavior (average of 1.8 illegal moves per legal move) [3, 1]. This indicates that the agent has learned the mechanics and informational role of illegal moves, even if strategic exploitation remains limited.

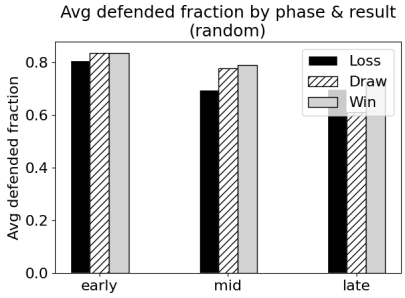

Fig. 1.12 The average percentage of defended pieces compared to the game phase

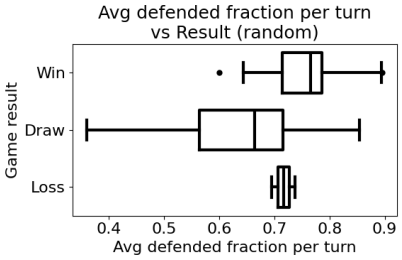

Fig. 1.13 The game result compared to the average percentage of pieces defended

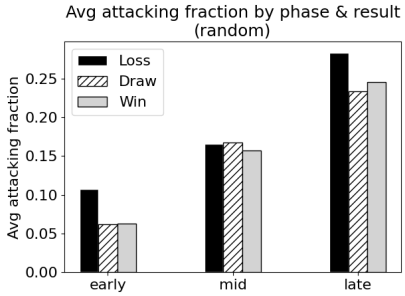

Fig. 1.14 The average percentage of pieces in potential attacking positions compared to the game phase

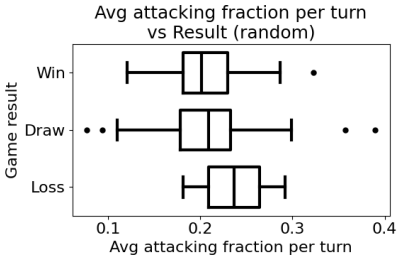

Fig. 1.15 The game result compared to the average percentage of pieces in potential attacking positions

When it comes to keeping pieces defended Fig. 1.12 shows that across all game phases, when you average the percentage of defended pieces, the agent model has learned very well to keep its pieces defended throughout the game.

When you then look at Fig. 1.13 you see as one might expect in a game of Kriegspiel that the winning games were all biased towards having strong defensive positions, with the games lost having worse defensive play, and the draws seemingly indicating an agent with a confused board structure that wasn't always conducive to defensive play.

Figure 1.14 shows the average percentage of pieces with available attacks across game phases. As expected, attacking opportunities generally increase as the game progresses and the board develops. However, games the agent ultimately lost exhibit a noticeably higher percentage of available attacks in both the early and late phases compared to wins or draws.

One plausible explanation, consistent with the illegal-move trends observed in Fig. 1.10, is deeper-than-expected opponent penetration into the agent's territory, increasing exposure and creating additional attack opportunities. This pattern per-

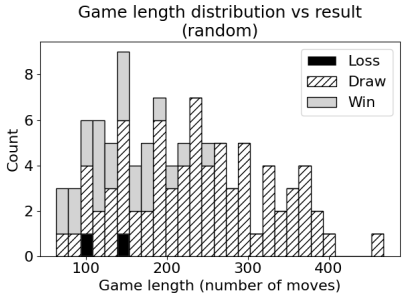

Fig. 1.16 Wins, draws, and losses, compared to game duration

sists in Fig. 1.15, where wins and draws show similar attack availability, while losses consistently exhibit elevated attacking potential.

Fig. 1.16 shows wins are clearly biased toward shorter games, while all games exceeding 260 moves result in draws. This pattern is consistent with the model's difficulty in acquiring sufficient board-state information in prolonged games, leading to increasingly inconsequential play and eventual termination under the 55-move draw rule.

1.5 Future Work

Validation loss trends in Fig. 1.3 indicate that, when properly integrated and tuned, fuzzy encoding can significantly improve training efficiency and representational quality. Future work will focus on translating these optimization benefits into full-game performance by redesigning the self-play reinforcement learning reward structure to overcome stagnation. Addressing long-horizon credit assignment and sparse endgame feedback is expected to substantially improve endgame execution and enable the hybrid fuzzy architecture to realize its full potential in competitive play.

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